Task 2: JavaScript Mini Application

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Student Name: Salva Akbar

Domain: Frontend Web Development

Objective

The objective of this task is to build a mini interactive JavaScript application using DOM manipulation.

In this project, a Simple Calculator was developed using HTML, CSS, and JavaScript. The calculator performs basic arithmetic operations such as addition, subtraction, multiplication, and division.

The application interacts with the Document Object Model (DOM) to update the display dynamically when the user clicks calculator buttons.

Tools & Technologies Used

The following technologies were used to develop this project:

- HTML5 – for structuring the application
- CSS3 – for styling the calculator interface
- JavaScript – for implementing functionality using DOM manipulation
- GitHub – for version control and code hosting
- GitHub Pages – for deploying the live project
- Visual Studio Code – for writing and editing code

Features of the Application

The calculator application includes the following features:

- Perform **addition**
- Perform **subtraction**
- Perform **multiplication**
- Perform **division**
- **Clear button** to reset the display
- Interactive calculator interface
- Simple and colorful user interface
- Uses **DOM manipulation to update the display**

Project Structure

The project folder contains the following files:

```
javascript-calculator
│
├── index.html
├── style.css
├── script.js
├── README.md
```

Code Snippets

HTML Structure (index.html):

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>JavaScript Calculator App</title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<div class="calculator">

<h2>Simple Calculator</h2>

<input type="text" id="display" disabled>

<div class="buttons">

<button onclick="appendValue('7')">7</button>

<button onclick="appendValue('8')">8</button>

<button onclick="appendValue('9')">9</button>

<button onclick="appendValue('/')">/</button>

<button onclick="appendValue('4')">4</button>

<button onclick="appendValue('5')">5</button>

<button onclick="appendValue('6')">6</button>
```

```
<button onclick="appendValue('*')">*</button>
<button onclick="appendValue('1')">1</button>
<button onclick="appendValue('2')">2</button>
<button onclick="appendValue('3')">3</button>
<button onclick="appendValue('-')">-</button>
<button onclick="appendValue('0')">0</button>
<button onclick="clearDisplay()">C</button>
<button onclick="calculate()">=</button>
<button onclick="appendValue('+')">+</button>
</div>
</div>
<script src="script.js"></script>
</body>
</html>
```

CSS Styling (style.css):

```
body{
font-family: Arial, sans-serif;
background: linear-gradient(135deg,#6a11cb,#2575fc);
display:flex;
justify-content:center;
align-items:center;
height:100vh;
margin:0;
}
.calculator{
background:white;
padding:25px;
border-radius:15px;
```

```
box-shadow:0 10px 25px rgba(0,0,0,0.3);
text-align:center;
}
#display{
width:100%;
height:45px;
margin-bottom:15px;
font-size:20px;
text-align:right;
padding:5px;
}
```

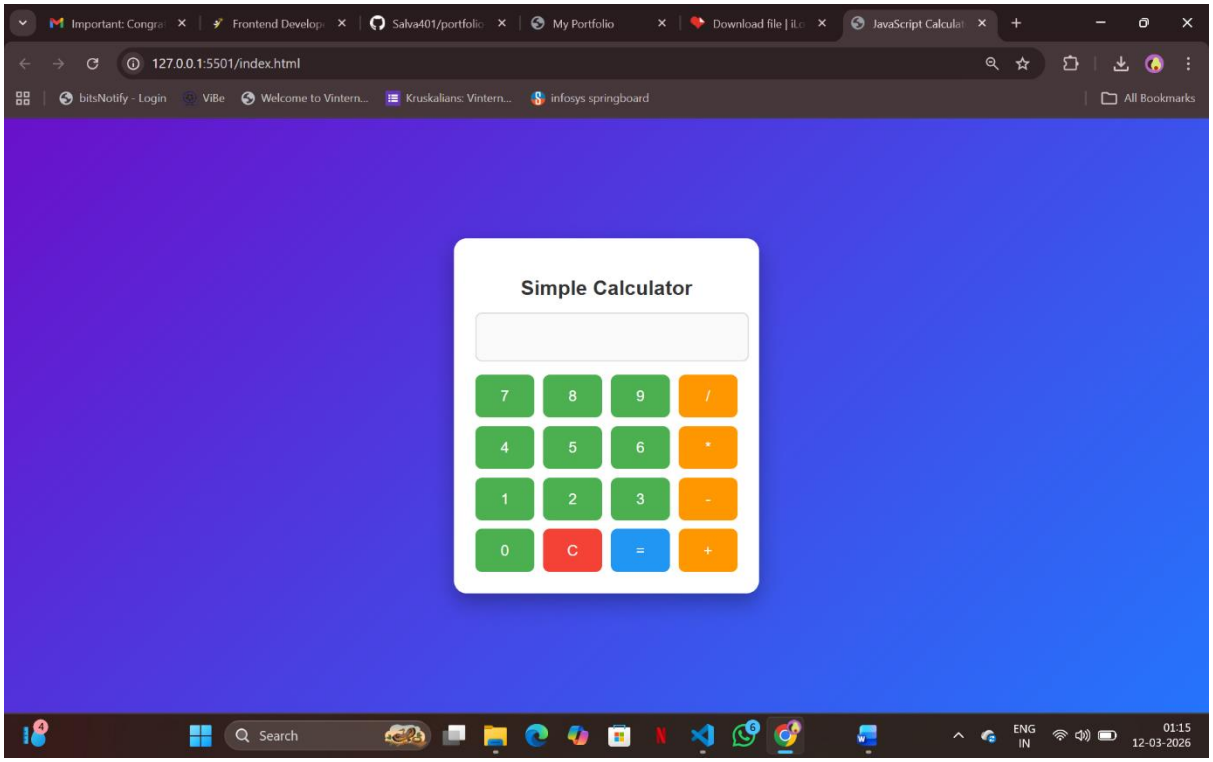
JavaScript Logic (script.js):

```
function appendValue(value){
document.getElementById("display").value += value;
}
function clearDisplay(){
document.getElementById("display").value = "";
}
function calculate(){
let result = document.getElementById("display").value;
try{
document.getElementById("display").value = eval(result);
}
catch{
alert("Invalid Calculation");
}
}
```

Screenshots

It Includes

- Calculator interface
- Mobile responsive view



GitHub Repository Link

Source code is available at: https://github.com/Salva401/Internspark-frontend-development-tasks/tree/main/Task2-JS-Calculator

Live Website Link

The project is hosted using GitHub Pages.

Live website: https://github.com/Salva401/Internspark-frontend-development-tasks/tree/main/Task2-JS-Calculator

Conclusion

This project demonstrates the use of **JavaScript DOM manipulation to create an interactive web application**. The calculator allows users to perform arithmetic operations dynamically using button clicks.

The project also demonstrates knowledge of **HTML structure, CSS styling, JavaScript functionality, and GitHub deployment**.

Author

Salva Akbar
Frontend Web Developer | Student